

$$\hat{Y} = Xw + \vec{1}b$$

$$Err = Y - \hat{Y}$$

$$E_{in} = \frac{1}{M} Err^T Err$$

$$\nabla_w E_{in} = -\frac{1}{M} X^T Err$$

$$\frac{dE_{in}}{db} = \frac{1}{M} \sum_{i=1}^n Err^{(i)}$$

$$w \leftarrow w - lr \nabla_w E_{in}$$

$$b \leftarrow b - lr \frac{dE_{in}}{db}$$

$b = \text{sesgo}$

$$\Theta^* = \arg \min_{\Theta} E_{in}(\Theta)$$

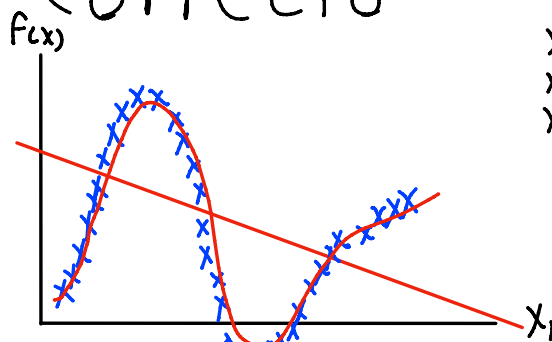
$$E_{in} \approx 0$$

$$f \approx h^* \rightarrow E_{in} \approx E_{out}$$

$$E_{out} \approx 0$$

Ingeniería de Características

Probablemente Aproximadamente Correcto

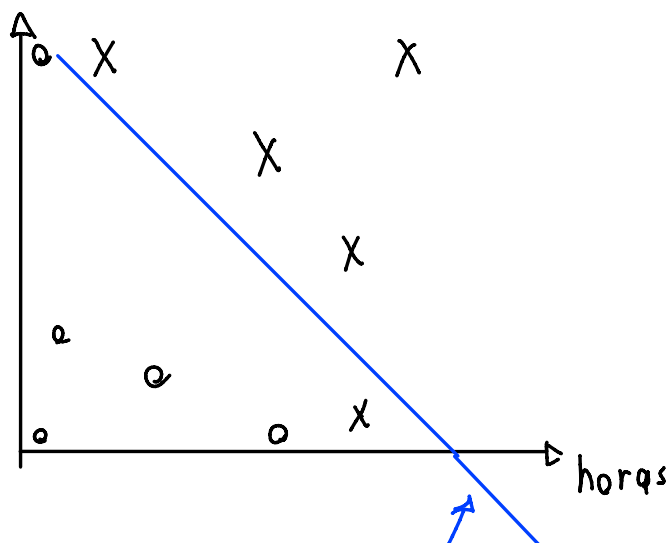


$$\begin{aligned} x_1 \\ x_2 = x_1^2 \\ x_3 = x_1^3 \end{aligned}$$

$$\begin{array}{cccc} x_1 & x_2 & x_3 & x_4 \\ x_1^{(1)} & x_2^{(1)} & x_3^{(1)} & y^{(1)} \\ \vdots & \vdots & \vdots & \vdots \\ x_1^{(m)} & x_2^{(m)} & x_3^{(m)} & y^{(m)} \end{array}$$

$$\hat{y} = w_1 x_1 + w_2 x_1^2 + w_3 x_1^3 + b$$

qsistencia



$$w_1 x_1 + w_2 x_2 + b = 0$$

generaliza bien

pero tiene un error grande

h_{Θ} = hipótesis en Θ

$$h_{\Theta} = \text{sign}(w_1 x_1 + w_2 x_2 + b)$$

$$X \subset \mathbb{R}^n$$

$$y \in \{h_{\Theta} \mid h_{\Theta} = \text{sign}(w^T x + b), w \in \mathbb{R}^n, b \in \mathbb{R}, \Theta = (w, b)\}$$

$$\text{loss}(y, \hat{y}) = \begin{cases} 1 & \text{si } y \neq \hat{y} \\ 0 & \text{en otro aspecto} \end{cases} = \max(-y\hat{y}, 0)$$

0/1-loss

$$E_{in}(w, b) = \frac{1}{M} \sum_{i=1}^M \text{loss}(y^{(i)}, \text{sign}(w^T x^{(i)} + b))$$

